Foundation for Virtual and Augmented Reality Systems

Prof. Samit Bhattacharya

Department of Computer Science and Engineering

Indian Institute of Technology, Guwahati

Week – 02

Lecture – 08

Lec 8: Issues & Challenges

Hello and welcome to the NPTEL MOOCs course on Foundations for Virtual and Augmented Reality Systems. We are going to start our next lecture, that is lecture number 5, which will be covering issues and challenges that are there in virtual and augmented reality systems. Before we start, like we were doing in the previous lectures, let us quickly recap what we have covered so far. So, we started our discussion with the basic understanding of the terms, broadly what they mean, what they refer to, and we also had a brief look at the importance, why this knowledge is important, why we should know about these terms, these technologies, primarily from the point of view of the commercial, potential applications and market caps. So, essentially from a commercial standpoint, we tried to understand why this is important. In the subsequent lectures, we went a little deeper. We explained the terms in terms of formal definitions of all the terms that we have come across. namely virtual reality, augmented reality and mixed reality.

Also, you may recollect that we used the term augmented reality; we decided to use the terms augmented reality and mixed reality as synonyms. So, they are used interchangeably. which we are going to follow in all our discussions in subsequent lectures. Then we covered again in brief some major application areas. While we are talking about application areas, application domains of individual technologies, we also had seen the characteristics, key characteristics, key defining characteristics we can say of each of these technologies. Now, these defining characteristics could help us to distinguish between say a virtual reality system and a non-virtual reality system or say an augmented reality system or a non-augmented reality system. Not only that, along with these types of distinctions, we are also able to differentiate between a virtual reality

system and an augmented reality system if we know those characteristics properly. So, we covered a few of the concepts in detail during those discussions.

In the last lecture, we learned about the historical evolution of the field, both virtual reality and augmented reality. We covered key milestones that occurred during this journey from the very beginning to the current state of the art. That is what we have covered so far. Now, in this lecture, what we are going to do is we are going to talk about some issues and challenges while dealing with such systems, as is the convention that we are following.

So, we will start our discussion on issues and challenges for virtual reality systems and in subsequent parts of the lecture, we are going to cover the same thing for augmented reality systems. So, let us begin our discussion on the key issues and challenges in virtual reality systems. or for developing or working with virtual reality systems. In order to understand the issues and challenges, let us quickly recollect the definition of virtual reality. We will be referring to the dictionary definition that we used in our earlier discussion on virtual reality.

What definition did we use there? Let us quickly recollect. A virtual reality system is an artificial environment which is experienced through sensory stimuli. Note that here we are referring to only two sensory organs to understand sight and to perceive sight and sound, but in reality we can actually for full immersion we need to use all our sensory organs. So, an artificial environment which is experienced through sensory stimuli provided by a computer and in which one's actions partially determine what happens in the environment.

So, earlier we went through this definition and we broke it down into key characteristics of a virtual reality system. So, no need to go into that discussion any further. However, what we can say is that based on this definition, we can say what is the objective, what is the main objective of any VR system. So, the objective is to let the user experience an artificial environment through sensory stimuli provided by the computer. So, here we are of course extending the definition a little bit instead of only sight and sound we are saying all sensory organs we are referring to all sensory organs. So, the overall objective of any VR system we can say is to let the user experience the artificial environment through our sensory organs or through the stimuli provided by the computer. That is the objective.

Now, it may appear to be very simple. However, this objective actually refers to two important things which are nothing but the main issues in experiencing VR. So, there are broadly two major issues in experiencing VR. One is the technological.

So, we said that we should be able to experience it through sensory stimuli provided by a computer. So, the technological issue is how to generate suitable stimuli to experience the virtual environment or artificial environment, but it does not end there. So, there is one more issue to be considered. What is that? That is psychological.

Now, we are saying that through technology, somehow we manage to generate some stimuli, suitable stimuli in other words. Now, how to perceive that stimuli suitably? So, that is the psychological issue. Technology has nothing to do with it. So, essentially this refers to the role of the human factor, how our psychology is made up, our psychological characteristics. Of course, as you can understand, different people will have different characteristics. So this perception of stimuli is likely to be different for different users. So to sum up, there are two major issues. One is technological: how to generate stimuli, or rather how to generate suitable stimuli; and one is psychological: how to perceive the stimuli suitably.

These two major issues in turn bring us to two major concerns, two important concerns while dealing with virtual reality system, technology, or experience. These are concerns that you will come across in almost every literature that deals with such technology. One is called immersion; the other one is called presence. Now, let us try to understand what these terms mean. Again, we will go through some definitions and we will try to understand based on the definitions what these terms mean.

Let us begin our discussion with immersion. What does immersion mean? Like we have done before while defining virtual reality, let us similarly begin with a definition of immersion. It is not a dictionary definition; rather, this is a definition taken from the literature which says: Immersion is the objective degree to which a VR system and application projects stimuli onto the sensory receptors of users in a way that is extensive, matching, surrounding, vivid, interactive and plot informing. So, what does it say? It says that immersion is the objective degree. A degree of what? Objective degree to which a VR system and application projects stimuli onto our or the human user's sensory receptors in a way such that the experience is extensive, matching, surrounding, vivid, interactive and plot informing. This is a definition which was originally proposed by Slater and Wilbur. I will give you the reference at the end of this lecture. So, the terms to

note in the definition are objective degree, VR system and application, projects stimuli, sensory receptors of users and of course, the seven terms namely extensive, matching, surrounding, vivid, interactive and plot informing, these six terms. So, in a nutshell, we can say that immersion is the objective degree which decides the qualities of experience, That a user is likely to have while dealing with a VR system. And these qualities are encapsulated in terms of these six terms: extensive, matching, surrounding, vivid, interactive and plot informing. Let us try to understand what these qualitative terms mean.

So, let us begin with the first term, extensive. So, what does it mean? It means the range of sensory modalities that are presented to the user. The more sensory organs are stimulated by the stimuli provided by the computer, the more extensive the immersion is. That is what this term refers to.

What can be the sensory organs? So, we have visual organs, auditory organs, physical force perception, smell, taste, touch, and so on. So, when we are talking about immersive experience, the more sensory organs we are using, the more extensive the experience is. That is what this term means. Then comes the second term.

Matching. What does it mean? So, it refers to congruence between sensory modalities. For example, appropriate visual presentation corresponding to head motion and virtual representation of one's own body should be there. So, when you are moving, say, your head, something in your visual field should move instantly without any delay. Otherwise, you will not get the perception that you are actually moving your head or your field of view. And this type of match between the user's physical movement and the movement in the virtual world is called matching. So, matching should be there while you are walking in the virtual world through some movement in the physical world or changing your field of view as I just explained or say moving your hand in the physical world.

So, similar things which let you perceive that you are moving your hand should happen in the virtual world and so on and so forth. Then comes the third quality, that is, surrounding. It is the extent to which cues are panoramic. The extent to which cues are panoramic. So, that is the meaning of this term, surrounding.

In other words, the extent of, say, when we are talking of visual senses, extent of field of view, When we are talking of auditory senses, the extent of our perception about auditory source and so on. So, wide field of view as compared to narrow field of view refers to more surrounding experience as compared to less surrounding experience. So, when I use

the term wide field of view, I am essentially referring to the quality, more surrounding as compared to when I am using the term narrow field of view.

So, that refers to the quality, less surrounding. Then vivid, what does it mean? The quality of energy stimulated within a particular modality. So, earlier we talked about using different modalities. Now, the vivid quality refers to the fact that within each of those modalities how much energy we are stimulating that means how much energy the computer is generating to stimulate that sensory organ. So, this quality is related to factors such as resolution, lighting, frame rate, audio bit rate; the more the better, as is obvious. So, if we have more resolution, then our visual perception would be better. If we have more audio bitrate, then our auditory perception will be better. If we have more lighting conditions, varied lighting conditions, brighter lights, then our visual perception will be better.

So, vividness means these types of things that enhance our perceptions through each of the sensory modalities. In other words, we can say that more resolution means more vivid immersion, more bit rate means more vivid immersion, and so on. Then comes the quality of interactivity. So when I say interaction or interactive quality, what does it mean?

Among these six, this is probably the most obvious one, that is the capability of the user to make changes to the world, response of virtual entities to user actions, and the user's ability to influence future events. Interactivity in a nutshell refers to Our ability, or rather the user's ability to interact with the virtual world, is not only one-way interaction to understand the response, but also the ability to change future events, the ability to influence future events.

So, if you recollect in our definition of VR, we said one of the defining characteristics of any VR system is interactivity. So, that is also reflected here. Finally, come to the last quality, which is plot informing. Now, this is very interesting. So, what this quality refers to is that through our interaction or other qualities, whether we are getting a feel of how things are happening, how things will happen in the future. I mean the consequences of our actions, whether we are able to understand, comprehend, or predict the consequences. In other words, it refers to the story, the story of the entire experience. So, here it is assumed that the experience is nothing but a story, a scripted story.

As the scripts, as the story, I mean, each of the instances changes through user interactions, and the story unfolds. The plot informing quality refers to our ability to

understand the overall story, not only the current instance of what is happening, but the overall story, the overall script. The more we are able to apprehend the script, the story, the more plot-informing the experience is. So, it refers to the story, which, in other words, refers to the consistent portrayal of a message or experience, dynamic unfolding of sequences of events, behavior of the world and its entities, whether the user is able to understand, comprehend, or predict those things, that is collectively we can say is referring to the understanding of the story. So, the more we understand the story, the more plot informing it is; the less we understand the story, the less plot informing the experience is. Now, that was about immersion, which is the technological aspect or technological issue for VR. So, just to summarize, there are two issues: technological and psychological. And correspondingly, there are two terms we refer to: immersion and presence.

Immersion refers to the technological issue. When we say we want to immerse or we want to generate, an immersive experience for the user, essentially we are referring to the technological side or the systemic side, which is captured in terms of these six qualities. And immersion refers to the objective degree to which these qualities are imparted to the user. And these qualities refer to activation of one or more sensory organs in a suitable way.

Now, that is the what technology can do. Let us come to the psychological side, that is presence. We may generate immersive qualities, but whether we will be able to perceive that or not, or whether the user will be able to perceive that or not, that is another issue which is captured by this term presence.

What is the definition, says? So, this definition is taken from the International Society for Presence Research. It was there since 2000. What it says is, presence is a psychological state or subjective perception in which even though part or all of an individual's current experience is generated by and/or filtered through human-made technology, part or all of the individual's perception fails to accurately acknowledge the role of the technology in the experience.

So, it is a little bit complicated. Let me go through it again. A psychological state or subjective perception, that is the first thing to note, in which even though part or all of an individual's current experience is, so whatever the individual is currently experiencing, either partly or fully, is generated by or filtered through human-made technology; part or all of the individual's perception, so whatever we are perceiving at that moment, fails to

accurately acknowledge fails to accurately acknowledge, note this term, the role of the technology in the experience. So, essentially what it means is that it is a psychological state, number one. Number two, in this state, we are getting input from technology, from computers, which is stimulating our sensory organs. So, we are perceiving something.

However, presence means that our perception should be as good as perceiving reality. That means we should not be able to distinguish between our perception of stimuli generated by a computer vis-à-vis our perception of stimuli generated from the real world. We should not be able to distinguish between the two. If that type of perception is happening, then we say that we are having a presence in the environment.

So, in other words, we can say that the computer is generating stimuli which are stimulating our sensory organs, but we fail to acknowledge that that these are generated by a computer; instead, we perceive them to be coming from the real world. So, in other words, we are perceiving that we are already in the real world; nothing changed. That means we are unable to distinguish between the virtual world and the real world; that is presence, or rather, to say, perfect presence. How are these two related: immersion and presence?

Immersion can be an objective and quantifiable description of what any particular system can provide or does provide, whereas presence is a state of consciousness or the psychological sense of being in the virtual environment. So, immersion is what the technology gives us. We can measure or we can comprehend what the technology can provide.

However, when we are talking about presence, it is inside the user's mind. There is no direct manifestation of it. It is difficult to objectively measure. So, subjective experience has to be taken into account. That is the major difference between these two concepts: immersion and presence.

Now, as is probably obvious by now, presence can be a function of both the user, that is the subjective experience, as well as immersion, that is the objective or technological experience. So, immersion aids presence, but it does not mean that if we provide immersive technology, then presence will automatically happen. Immersion is capable of producing presence, but not necessarily always. Say, for example, you are asking a user to experience a virtual reality system.

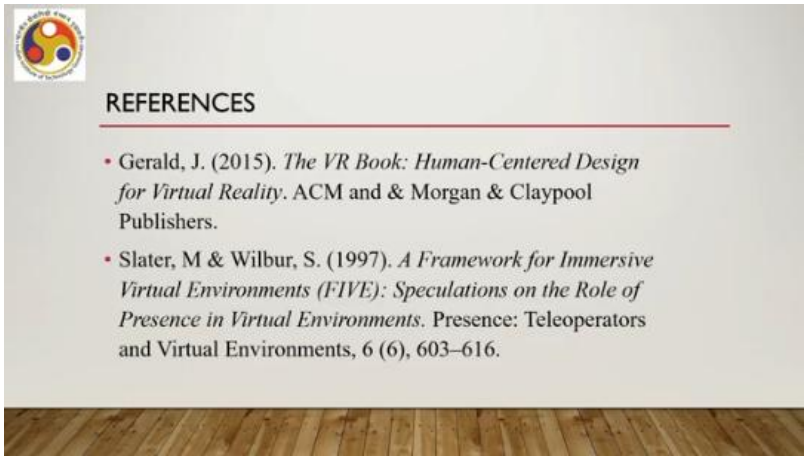
You ask the user to wear the headset and do all the things, all the interactions that are supported by the system. But if the user simply keeps the eyes shut, then and imagines that he or she is in some other place, then all these technological things will come to naught. It will mean nothing. So, the user can simply shut their eyes and cut themselves off from that virtual experience.

So, immersion is still there, but presence will not be there. On the other hand, presence can be limited by immersion. The greater immersion a system or application provides, the greater potential for a user to feel present in that virtual world. The lesser the immersive technology provides, the lesser is the presence. That is there.

However, as we have just seen, immersion does not ensure presence, but immersion can aid enhanced perception of presence. So with this knowledge, we can say that the overall goal of any virtual reality system is to create a high degree of presence, or in other words, to make users feel that they are really in the virtual world, but they are unable to distinguish between the real world and the virtual world. So earlier we said the goal of a VR system is to give an immersive experience. Now we are refining and saying that the goal is to create a high degree of presence because now we know what presence is and what immersion is.

Now to create a high degree of presence, we have to have a system that provides a high degree of immersion, and that is a systemic issue. So with that, we have come to the end of this part of the lecture where we have covered a few important concepts, namely the technological challenge and the concept of immersion, the psychological challenge and the concept of presence and how they are related, how they are different, what are the things to be considered while creating an immersive experience and what is the overall goal of any VR system. But that is not all; immersion and presence are not the only issues that are afflicting any VR experience.

There are other issues as well, and there are issues for AR systems also. Those we will be taking up in the next part of the lecture. And these are the references from where those terms were taken and definitions were taken. If you go through them, you will get to know more about these concepts. The second one, that is Slater and Wilbur, 1997.



This is one of the earliest papers related to this topic. And I strongly suggest that you go through this reference to get a better understanding of how things were dealt with at that point in time. So, with that, we have come to the end of this lecture. I hope you have enjoyed it. We will cover the other things related to issues and challenges in the next part.

Until then, goodbye. Thank you, and happy learning.